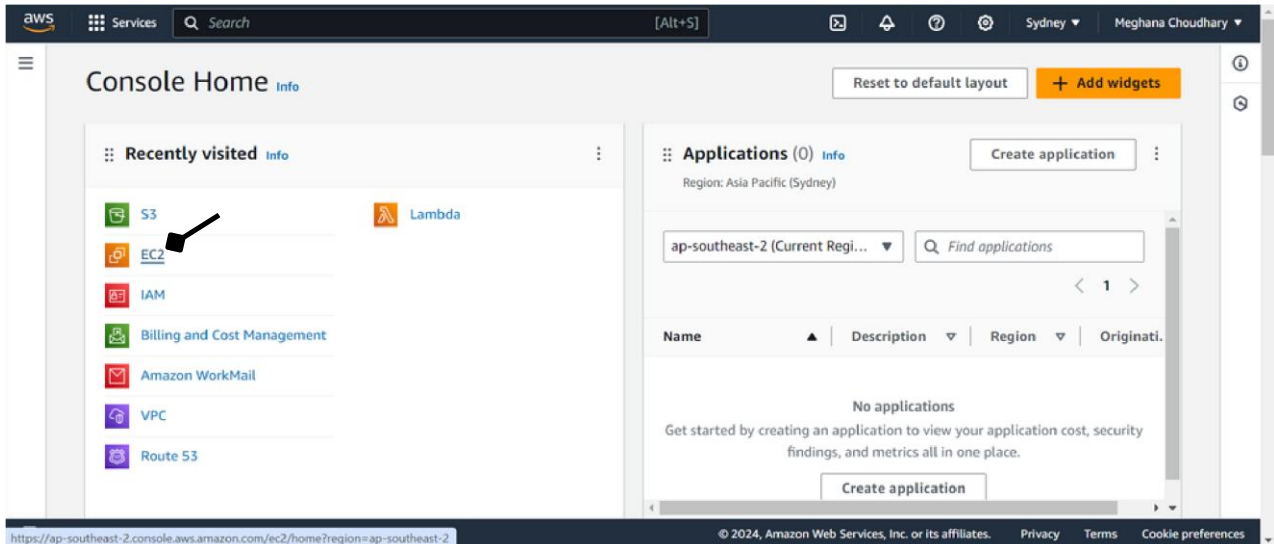


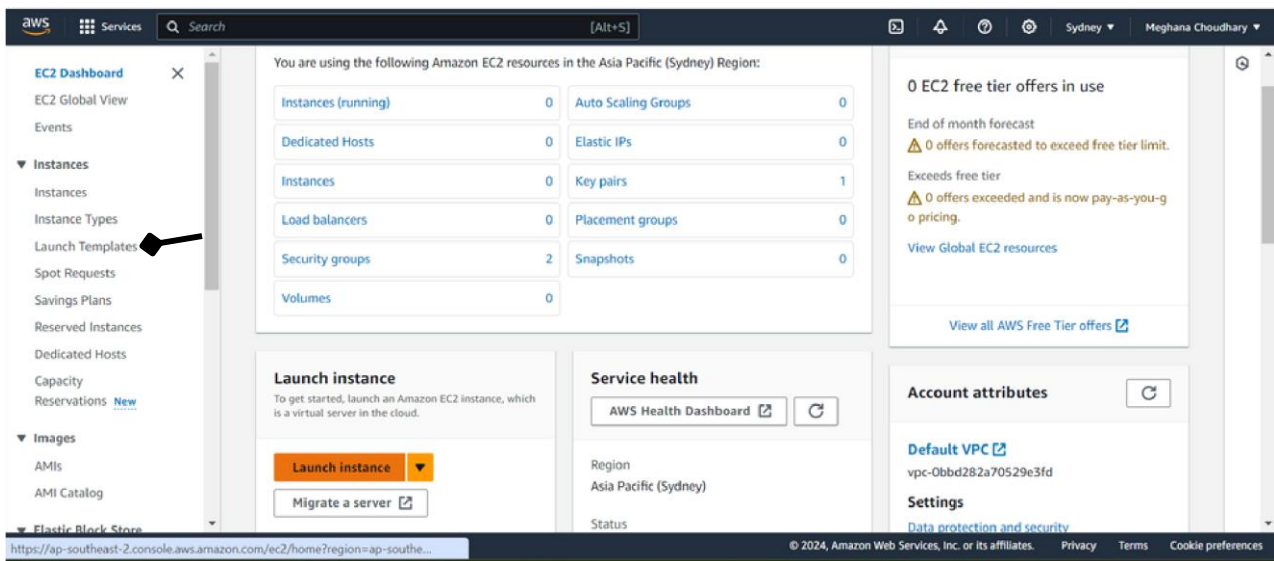
Assignment-11

PROBLEM STATEMENT: Build scaling plans in AWS that balance load on different EC2 instances.

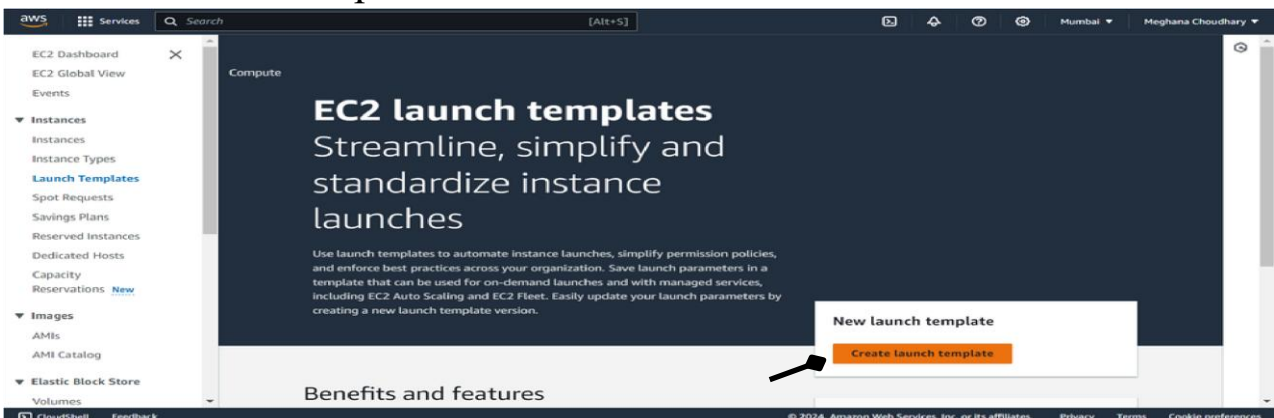
1. Log into AWS console and click on "EC2".



2. Click on 'Launch Templates'.



3. Create launch templates.



4. Enter necessary details. In 'Launch template name and description' give a name and description to your launch template and select the check box for 'Provide guidance to help me set up a template that I can use with EC2 Auto Scaling'

aws Services Search [Option+S] Mumbai Meghana Choudhary

EC2 > Launch templates > Create launch template

Create launch template

Creating a launch template allows you to create a saved instance configuration that can be reused, shared and launched at a later time. Templates can have multiple versions.

Launch template name and description

Launch template name - *required*

Must be unique to this account. Max 128 chars. No spaces or special characters like '&', "'", '@'.

Template version description

Max 255 chars

Auto Scaling guidance [Info](#)

Select this if you intend to use this template with EC2 Auto Scaling

☒ Provide guidance to help me set up a template that I can use with EC2 Auto Scaling

▶ Template tags

▶ Source template

Summary

Software Image (AMI)

-

Virtual server type (instance type)

-

Firewall (security group)

-

Storage (volumes)

-

Free tier: In your first year includes 750 hours of t2.micro (or t3.micro in the Regions in which t2.micro is unavailable) instance usage on free tier AMIs per month, 750 hours of public IPv4 address usage per month, 30 GiB

Cancel **Create launch template**

CloudShell Feedback © 2024, Amazon Web Services, Inc. or its affiliates. Privacy Terms Cookie preferences

For OS Image select 'Ubuntu Server'

aws Services Search [Option+S] Mumbai Meghana Choudhary

Launch template contents

Specify the details of your launch template below. Leaving a field blank will result in the field not being included in the launch template.

Application and OS Images (Amazon Machine Image) [Info](#)

An AMI is a template that contains the software configuration (operating system, application server, and applications) required to launch your instance. Search or Browse for AMIs if you don't see what you are looking for below

Search our full catalog including 1000s of application and OS images

Recents Quick Start

Don't include in launch template

Amazon Linux

macOS

Ubuntu

Windows

[Browse more AMIs](#)

Including AMIs from AWS, Marketplace and the Community

Amazon Machine Image (AMI)

Ubuntu Server 24.04 LTS (HVM), SSD Volume Type

ami-0f58b397bc5c1f2e8 (64-bit (x86)) / ami-0dda4ba9a42839a4b (64-bit (Arm))

Virtualization: hvm ENA enabled: true Root device type: ebs

Free tier eligible

Description

Canonical, Ubuntu, 24.04 LTS, amd64 noble image build on 2024-04-23

Architecture

64-bit (x86)

AMI ID

ami-0f58b397bc5c1f2e8

Verified provider

Summary

t2.micro

Firewall (security group)

:4000 open

Storage (volumes)

1 volume(s) - 8 GiB

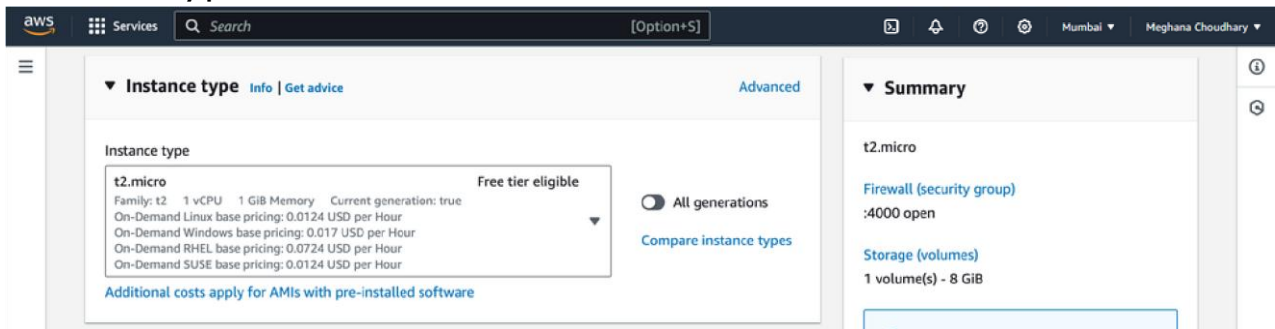
Free tier: In your first year includes 750 hours of t2.micro (or t3.micro in the Regions in which t2.micro is unavailable) instance usage on free tier AMIs per month, 750 hours of public IPv4 address usage per month, 30 GiB of EBS storage, 2 million IOs, 1 GB of snapshots, and 100 GB of bandwidth to the internet.

t2.micro is unavailable) instance usage on free tier AMIs per month, 750 hours of public IPv4 address usage per month, 30 GiB of EBS storage, 2 million IOs, 1 GB of snapshots, and 100 GB of bandwidth to the internet.

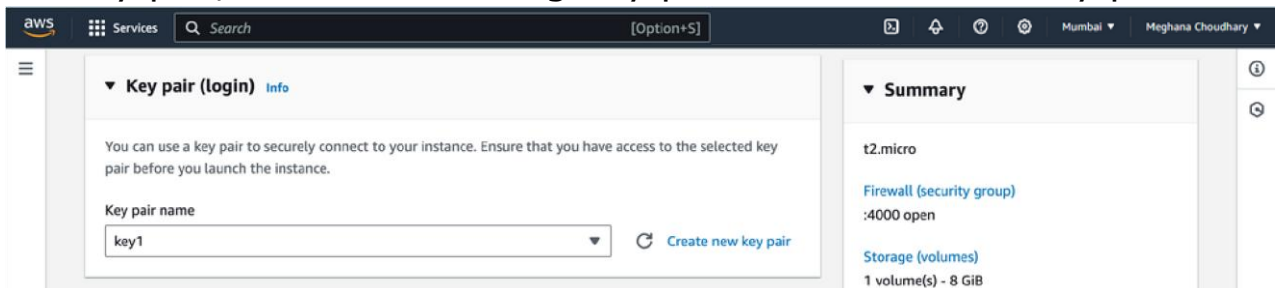
Cancel **Create launch template**

CloudShell Feedback © 2024, Amazon Web Services, Inc. or its affiliates. Privacy Terms Cookie preferences

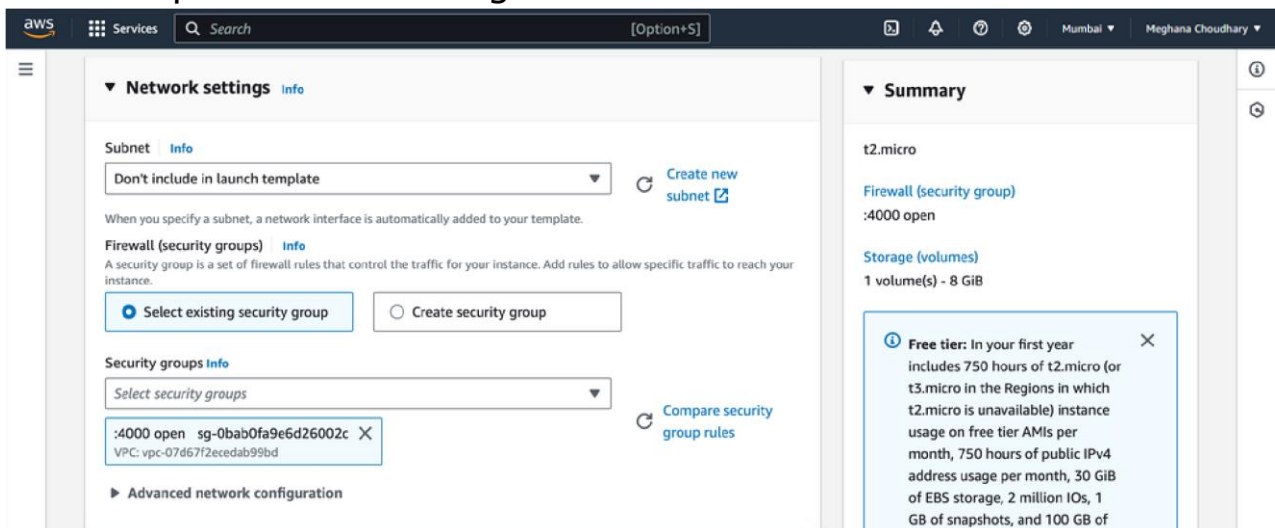
For instance type, select 't2.micro' or any other free tier eligible instance type.



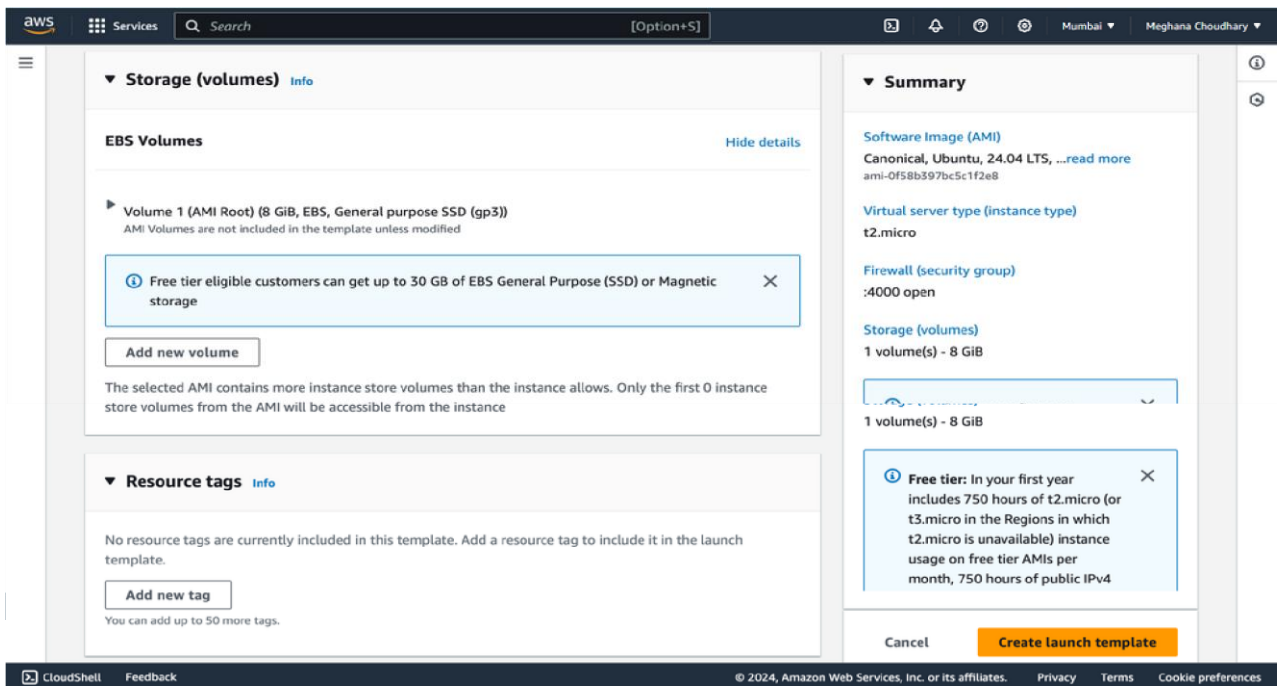
For key pair, select an existing key pair or create a new key pair.



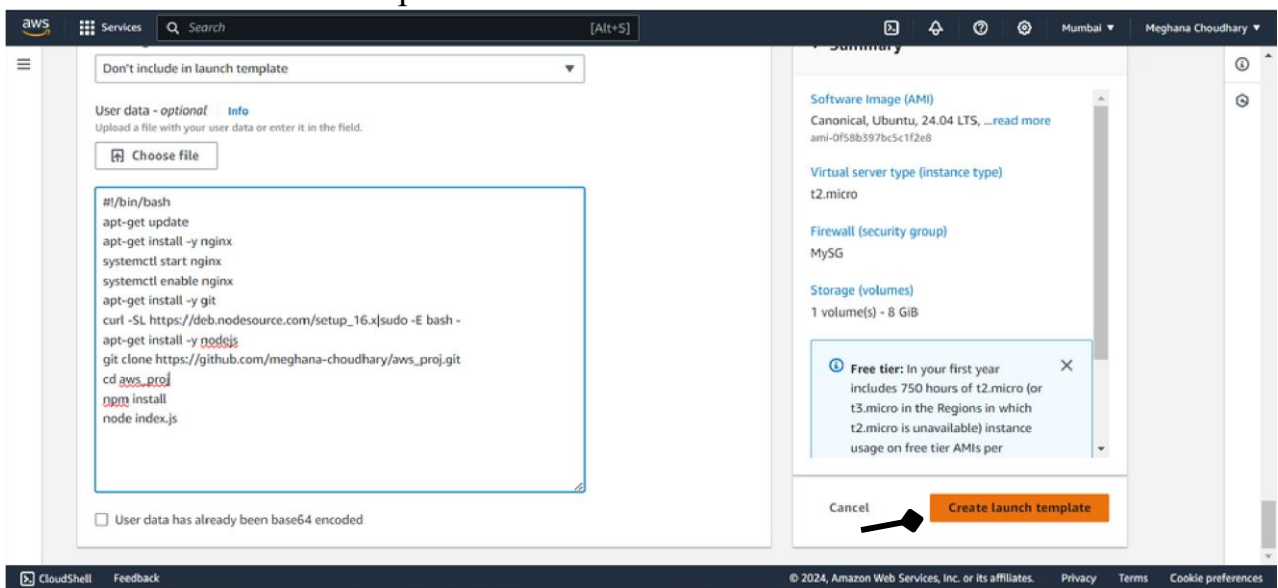
For security group select an existing security group where port 4000 is open for incoming traffic.



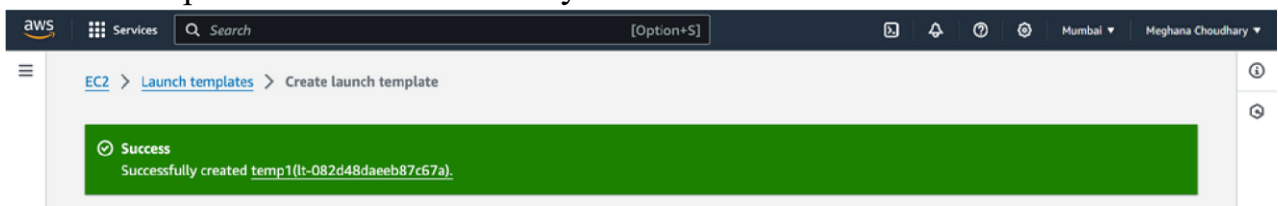
Keep Storage and Resource Tag as default.



In Advanced details scroll down to the bottom and add the text to the user data. Then click on Create launch template.



Launch Templates created successfully.



5. Scroll down and click on View launch templates.

Next Steps

Launch an instance
With On-Demand Instances, you pay for compute capacity by the second (for Linux, with a minimum of 60 seconds) or by the hour (for all other operating systems) with no long-term commitments or upfront payments. Launch an On-Demand Instance from your launch template.
[Launch instance from this template](#)

Create an Auto Scaling group from your template
Amazon EC2 Auto Scaling helps you maintain application availability and allows you to scale your Amazon EC2 capacity up or down automatically according to conditions you define. You can use Auto Scaling to help ensure that you are running your desired number of Amazon EC2 instances during demand spikes to maintain performance and decrease capacity during lulls to reduce costs.
[Create Auto Scaling group](#)

Create Spot Fleet
A Spot Instance is an unused EC2 instance that is available for less than the On-Demand price. Because Spot Instances enable you to request unused EC2 instances at steep discounts, you can lower your Amazon EC2 costs significantly. The hourly price for a Spot Instance (of each instance type in each Availability Zone) is set by Amazon EC2, and adjusted gradually based on the long-term supply of and demand for Spot Instances. Spot instances are well-suited for data-analysis, batch jobs, background processing, and optional tasks.
[Create Spot Fleet](#)

[View launch templates](#)

6. Now go to 'Auto Scaling Groups'

Launch Templates (1) [Info](#)

[Refresh](#) [Actions](#) [Create launch template](#)

	Launch Template ID	Launch Template Name	Default Version	Latest Version	Create Time
☐	lt-082d48daeeb87c67a	temp1	1	1	2024-05-20T...

Select a launch template

7. Add Details for the Auto Scaling Group.

Step 1: Choose launch templates- Give an name to the Auto Scaling Group and choose the Launch template you just created as the launch template.

Click on Next.

Step 2: Choose instance launch options- Choose all the ‘Availability Zones’ available.
Click on Next.

© 2024, Amazon Web Services, Inc. or its affiliates. [Privacy](#) [Terms](#) [Cookie preferences](#)

Step 3: Configure advanced options- For Load balancing select ‘Attach to a new load balancer’

aws Services Search [Option+S] Mumbai Meghana Choudhary

EC2 > Auto Scaling groups > Create Auto Scaling group

Step 1
[Choose launch template](#)

Step 2
[Choose instance launch options](#)

Step 3 - optional
Configure advanced options

Step 4 - optional
[Configure group size and scaling](#)

Step 5 - optional
[Add notifications](#)

Configure advanced options - optional Info

Integrate your Auto Scaling group with other services to distribute network traffic across multiple servers using a load balancer or to establish service-to-service communications using VPC Lattice. You can also set options that give you more control over health check replacements and monitoring.

Load balancing Info

Use the options below to attach your Auto Scaling group to an existing load balancer, or to a new load balancer that you define.

- ☐ No load balancer
Traffic to your Auto Scaling group will not be fronted by a load balancer.
- ☐ Attach to an existing load balancer
Choose from your existing load balancers.
- ☒ Attach to a new load balancer
Quickly create a basic load balancer to attach to your Auto Scaling group.

For new load balancer configuration select ‘Application Load Balancer’, ‘Internet-facing’ and make HTTP port from 80 to 4000 while creating a target group.

aws Services Search [Option+S] Mumbai Meghana Choudhary

Step 6 - optional
[Add tags](#)

Step 7
[Review](#)

Attach to a new load balancer

Define a new load balancer to create for attachment to this Auto Scaling group.

Load balancer type

Choose from the load balancer types offered below. Type selection cannot be changed after the load balancer is created. If you need a different type of load balancer than those offered here, [visit the Load Balancing console](#).

- ☒ Application Load Balancer
HTTP, HTTPS
- ☐ Network Load Balancer
TCP, UDP, TLS

Load balancer name

Name cannot be changed after the load balancer is created.

AutoScale1-1

Load balancer scheme

Scheme cannot be changed after the load balancer is created.

- ☐ Internal
- ☒ Internet-facing

Network mapping

Your new load balancer will be created using the same VPC and Availability Zone selections as your Auto Scaling group. You can select different subnets and add subnets from additional Availability Zones.

VPC
[vpc-07d67f2ecedab99bd](#)

Availability Zones and subnets

You must select a single subnet for each Availability Zone enabled. Only public subnets are available for selection to support DNS resolution.

☒ ap-south-1b	subnet-0c5adeb10a1552feb
☒ ap-south-1a	subnet-062f2f3a2babf4755
☒ ap-south-1c	subnet-0225191dd13ccd1ee

Listeners and routing

If you require secure listeners, or multiple listeners, you can configure them from the [Load Balancing console](#) after your load balancer is created.

Protocol	Port	Default routing (forward to)
HTTP	4000	Create a target group

New target group name
An instance target group with default settings will be created.

AutoScale1-1

Tags - optional

Consider adding tags to your load balancer. Tags enable you to categorize your AWS resources so you can more easily manage them.

[Add tag](#)

50 remaining

CloudShell Feedback © 2024, Amazon Web Services, Inc. or its affiliates. Privacy Terms Cookie preferences

Health checks tick the check box for ‘Turn on Elastic Load Balancing health checks’ and make the health check grace period 240 seconds.
Click on Next.

VPC Lattice integration options [info](#)

To improve networking capabilities and scalability, integrate your Auto Scaling group with VPC Lattice. VPC Lattice facilitates communications between AWS services and helps you connect and manage your applications across compute services in AWS.

Select VPC Lattice service to attach

☒ **No VPC Lattice service**
VPC Lattice will not manage your Auto Scaling group's network access and connectivity with other services.

☐ **Attach to VPC Lattice service**
Incoming requests associated with specified VPC Lattice target groups will be routed to your Auto Scaling group.

[Create new VPC Lattice service](#)

Health checks

Health checks increase availability by replacing unhealthy instances. When you use multiple health checks, all are evaluated, and if at least one fails, instance replacement occurs.

EC2 health checks

[Always enabled](#)

Additional health check types - optional [info](#)

☒ **Turn on Elastic Load Balancing health checks** **Recommended**
Elastic Load Balancing monitors whether instances are available to handle requests. When it reports an unhealthy instance, EC2 Auto Scaling can replace it on its next periodic check.

☐ **EC2 Auto Scaling will start to detect and act on health checks performed by Elastic Load Balancing. To avoid unexpected terminations, first verify the settings of these health checks in the [Load Balancer console](#)**

☐ **Turn on VPC Lattice health checks**
VPC Lattice can monitor whether instances are available to handle requests. If it considers a target as failed a health check, EC2 Auto Scaling replaces it after its next periodic check.

Health check grace period [info](#)

This time period delays the first health check until your instances finish initializing. It doesn't prevent an instance from terminating when placed into a non-running state.

seconds

Additional settings

Monitoring [info](#)

☐ **Enable group metrics collection within CloudWatch**

Default instance warmup [info](#)

The amount of time that CloudWatch metrics for new instances do not contribute to the group's aggregated instance metrics, as their usage data is not reliable yet.

☐ **Enable default instance warmup**

Cancel Skip to review Previous **Next**

Step 4: Configure group size and scaling- In Group size set Desired capacity as 2. In Scaling set Min desired capacity as 2 and Max desired capacity as 3. In Automatic scaling choose target scaling policy and set the instance warmup as 240 second. Keeping everything else as default, Click on Next.

aws Services Search [Option+S] Mumbai Meghana Choudhary

EC2 > Auto Scaling groups > Create Auto Scaling group

Step 1
[Choose launch template](#)

Step 2
[Choose instance launch options](#)

Step 3 - optional
[Configure advanced options](#)

Step 4 - optional
Configure group size and scaling

Step 5 - optional
[Add notifications](#)

Step 6 - optional
[Add tags](#)

Step 7
[Review](#)

Configure group size and scaling - optional

Define your group's desired capacity and scaling limits. You can optionally add automatic scaling to adjust the size of your group.

Group size

Set the initial size of the Auto Scaling group. After creating the group, you can change its size to meet demand, either manually or by using automatic scaling.

Desired capacity type
Choose the unit of measurement for the desired capacity value. vCPUs and Memory(GiB) are only supported for mixed instances groups configured with a set of instance attributes.
Units (number of instances)

Desired capacity
Specify your group size.
2

Scaling

You can resize your Auto Scaling group manually or automatically to meet changes in demand.

Scaling limits
Set limits on how much your desired capacity can be increased or decreased.

Min desired capacity
2
Equal or less than desired capacity

Max desired capacity
3
Equal or greater than desired capacity

Automatic scaling - optional
Choose whether to use a target tracking policy. You can set up other metric-based scaling policies and scheduled scaling after creating your Auto Scaling group.

☐ No scaling policies
Your Auto Scaling group will remain at its initial size and will not dynamically resize to meet demand.

☒ Target tracking scaling policy
Choose a CloudWatch metric and target value and let the scaling policy adjust the desired capacity in proportion to the metric's value.

Scaling policy name
Target Tracking Policy

Metric type
Monitored metric that determines if resource utilization is too low or high. If using EC2 metrics, consider enabling detailed monitoring for better scaling performance.
Average CPU utilization

Target value
50

Instance warmup
240 seconds

☐ Disable scale in to create only a scale-out policy

Skip through Step 5 and Step 6 by clicking Next.
Finally in Step 7 Click on Create Auto Scaling Group.

aws Services Search [Option+S] Mumbai Meghana Choudhary

Instance scale-in protection

Instance scale-in protection
☐ Enable instance protection from scale in

Step 5: Add notifications [Edit](#)

Notifications
No notifications

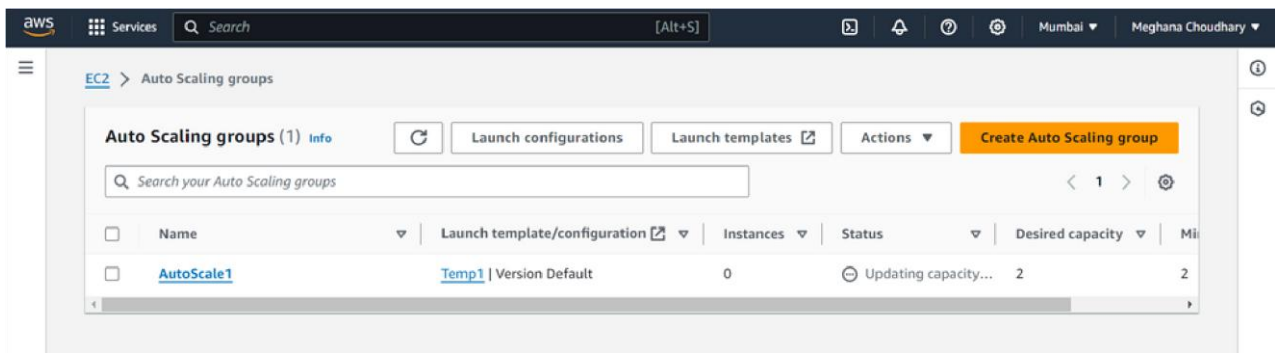
Step 6: Add tags [Edit](#)

Tags (0)

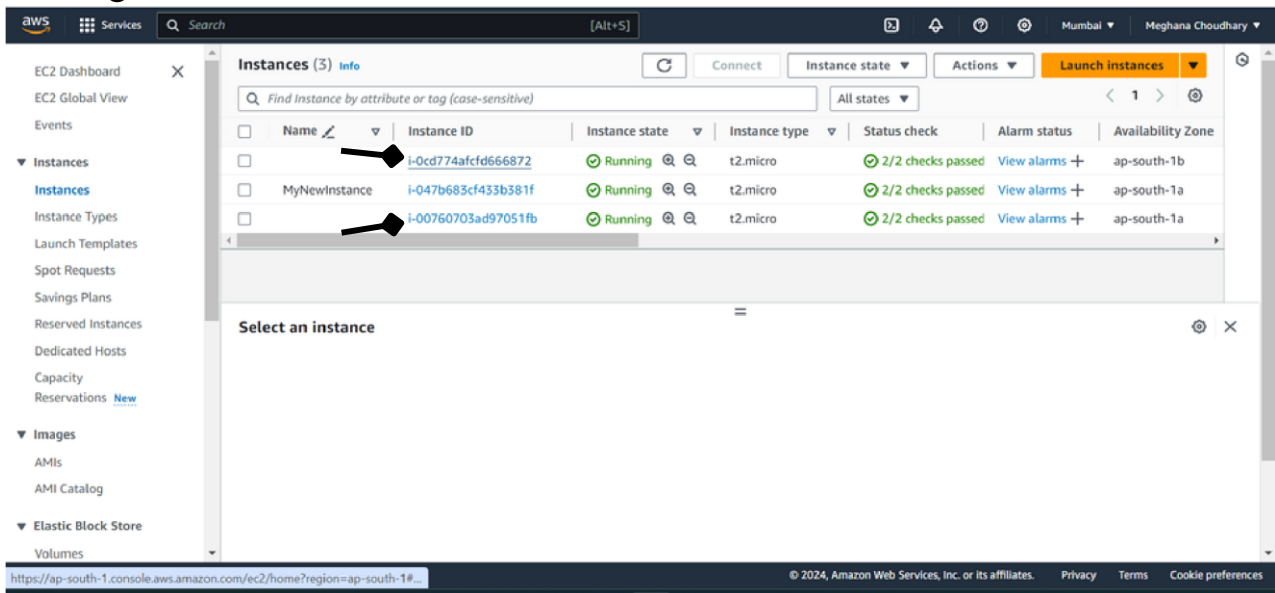
Key	Value	Tag new instances
No tags		

Cancel Previous **Create Auto Scaling group**

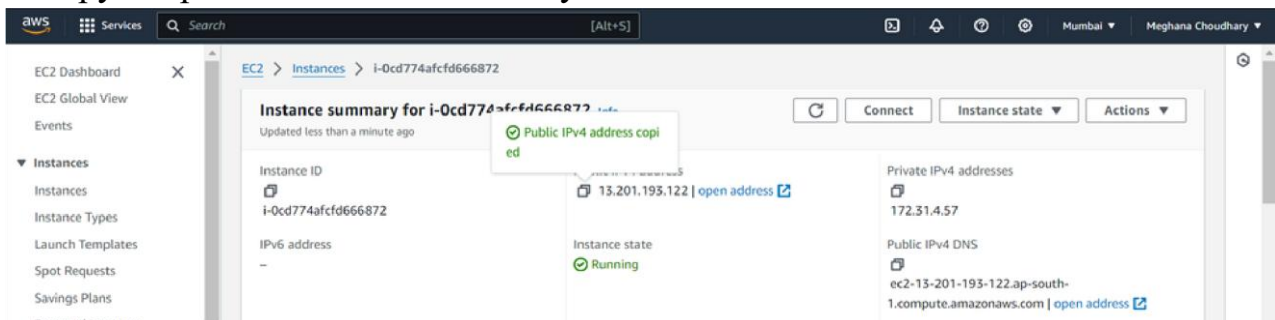
Auto Scaling Group is created.



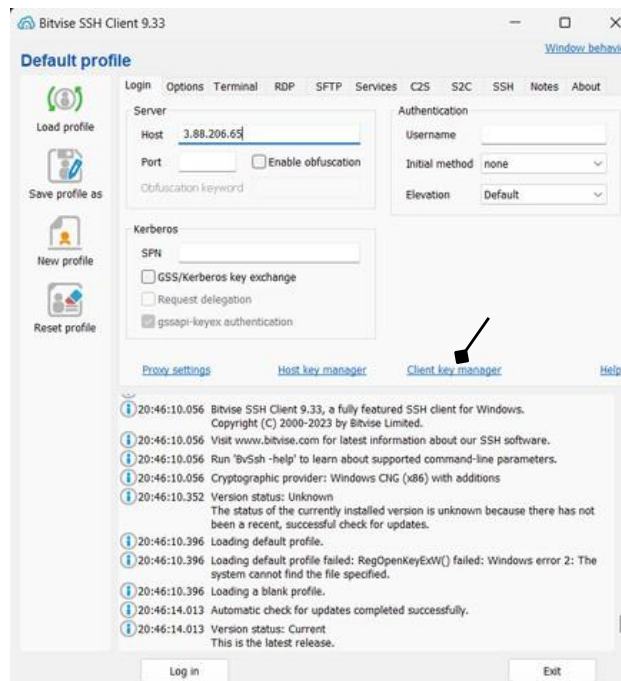
8. Going back to instances we can see two unnamed instances are created.



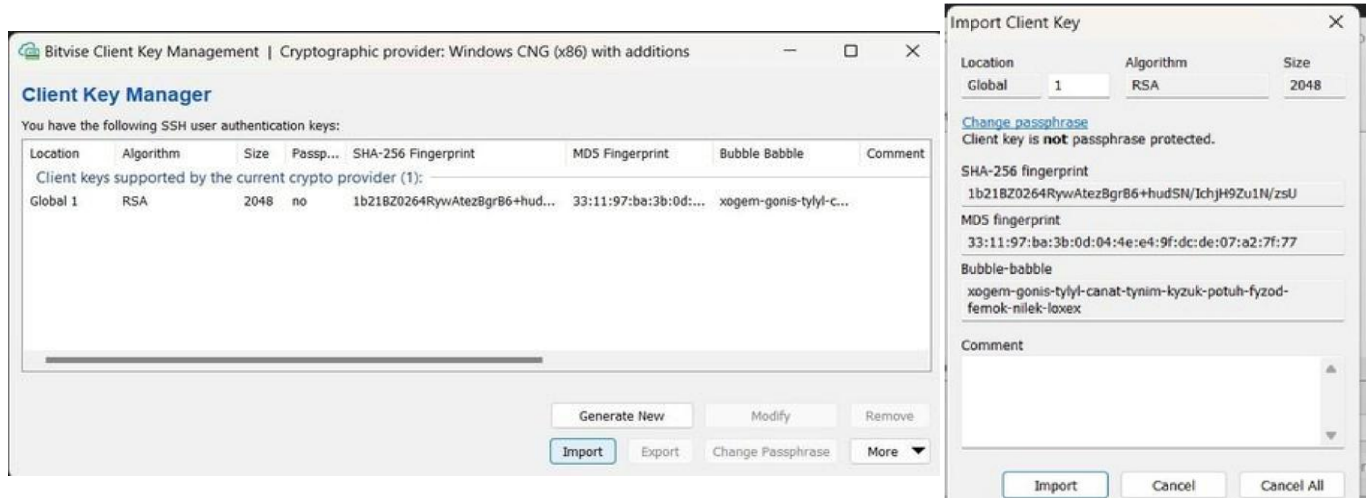
8. Copy the public IPv4 address of any one of the two instance.



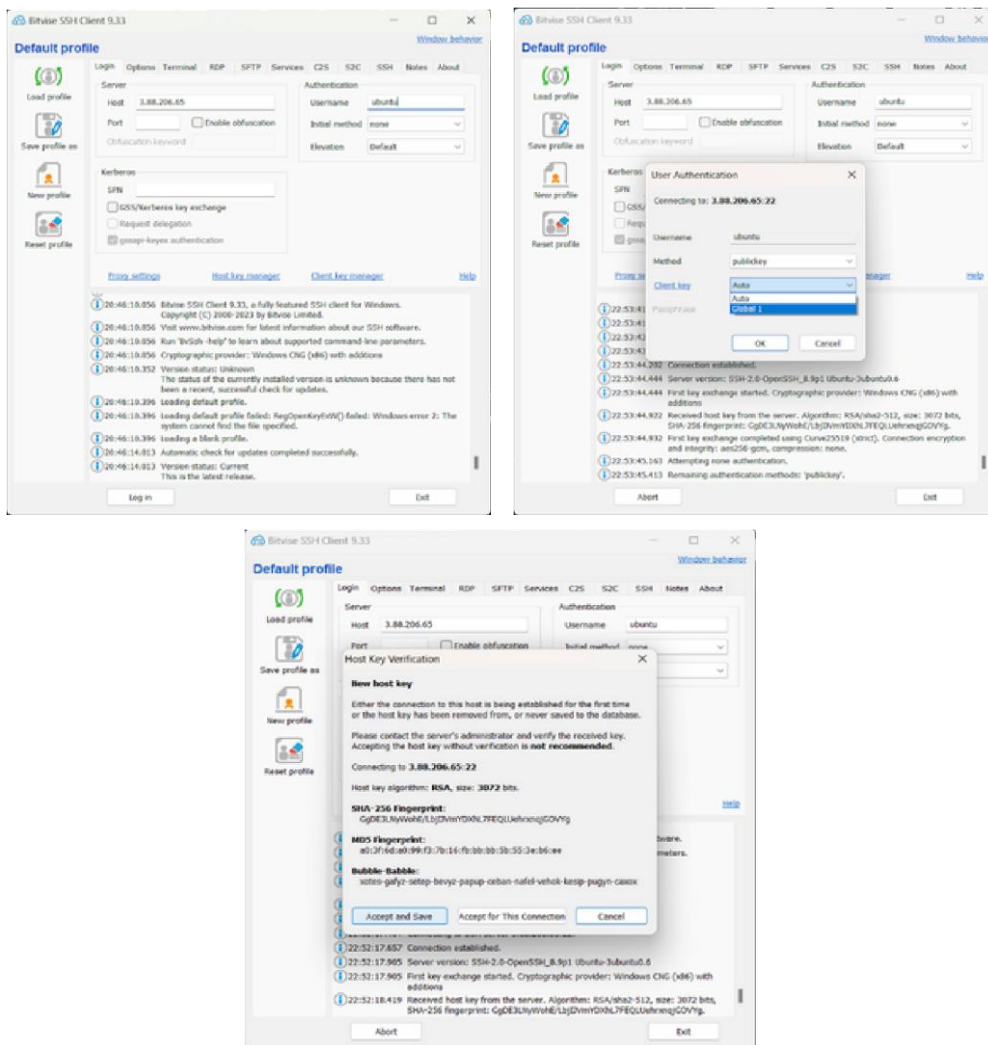
9. In Bitwise SSH Client, paste the 'Public IPV4 address' and click on 'Client keyManager'.



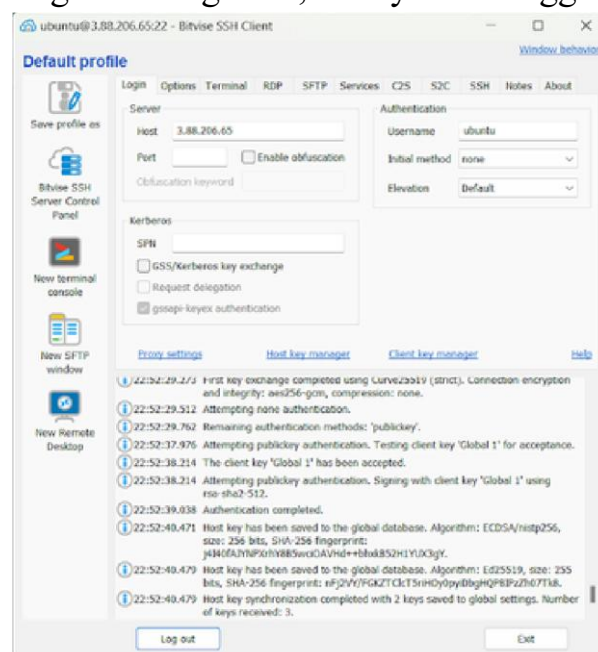
10. Under 'Client key manager', if there is any existing key then remove it and click on 'Import', select the same key that you used while creating the instance, from your file directory and then again click on 'Import' and then the key is successfully imported.



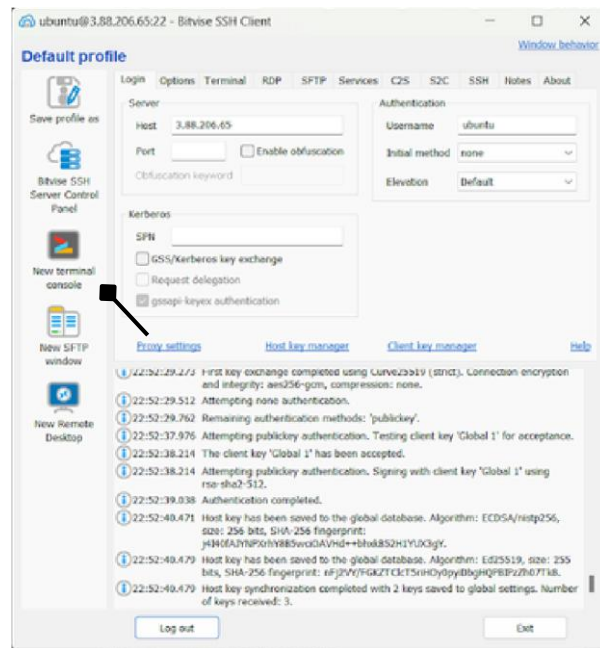
10. In 'Bitvise SSH Client', under 'Default profile' give the 'Username' as 'ubuntu' then select 'Global1' from 'Client Key Manager' then click on 'Log in' & 'Accept and save'.



If the 'Log In' button changes to 'Log Out', then you are logged in successfully.



11. Click on 'New terminal console', to access the terminal of your instance.



8. Once you are inside the terminal of your instance

To create a infinitely running bash script vim
infi.sh

```
ubuntu@ip-172-31-12-249:~$ vim infi.sh
```

In the bash script, add the following code.

```
#!/bin/bash
while(true) do
echo "Inside Loop"
done
```

```
#!/bin/bash

while(true)
do
    echo "InsideLoop"
done
```

To make the script executable sudo
chmod 777 infi.sh

```
ubuntu@ip-172-31-5-108:~$ sudo chmod 777 infi.sh
```

Run the script sh

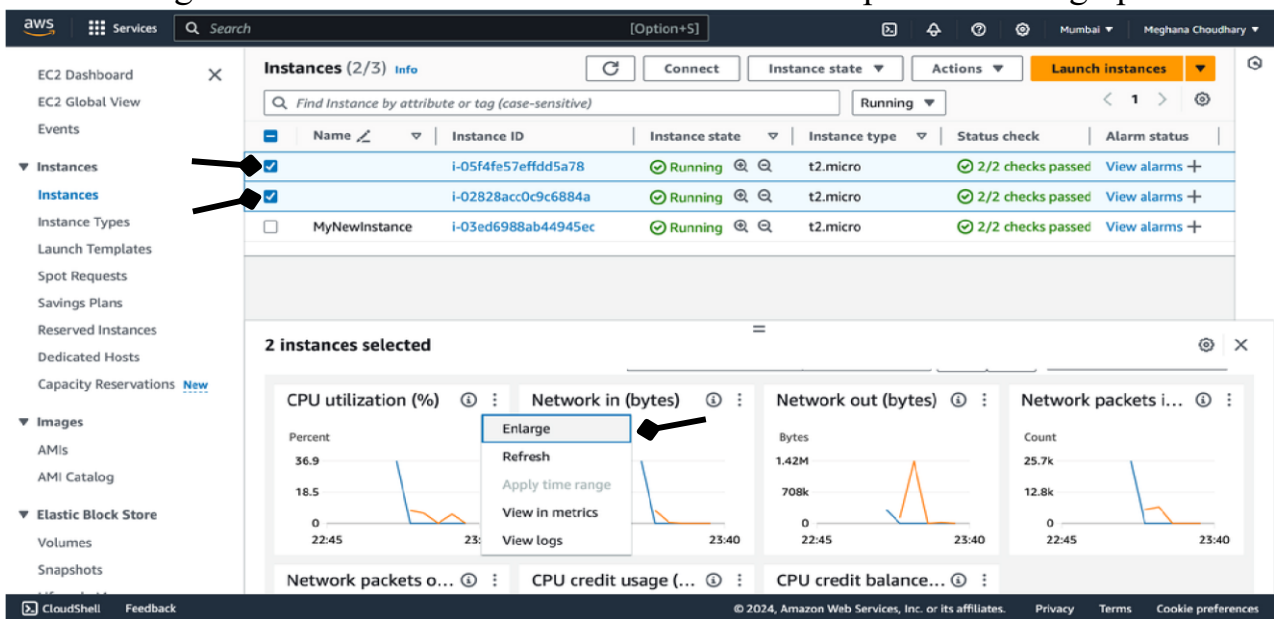
```
infi.sh
```

9. An infinite loop starts to run, stressing the cpu of the instance.

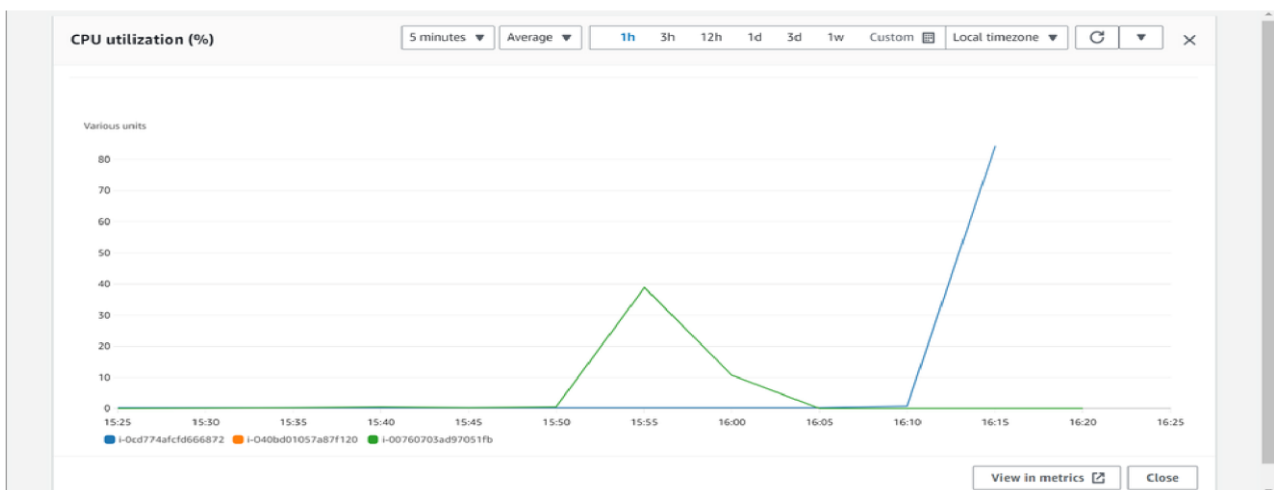
```
ubuntu@ip-172-31-5-108:~$ sh infi.sh
```


[illegible]

10. Selecting the two instances we can take a look at the cpu utilisation graph.



11. After a while the instance reaches and surpasses it's 50% CPU utilisationthreshold.



12. So to take that load off the instance, the auto scaler spins up another instance to balance the load.

aws

Services

Search

[Alt+S]

Mumbai

Meghana Choudhary

EC2 Dashboard

EC2 Global View

Events

Instances

Instances

Instance Types

Launch Templates

Spot Requests

Savings Plans

Reserved Instances

Dedicated Hosts

Capacity

Reservations

Images

AMIs

AMI Catalog

Elastic Block Store

Volumes

Instances (4)

Info

Find Instance by attribute or tag (case-sensitive)

Instance state = running

Clear filters

Connect

Instance state

Actions

Launch instances

All states

< 1 >

	Name	Instance ID	Instance state	Instance type	Status check	Alarm status	Availability Zone
☐		i-040bd01057a87f120	Running	t2.micro	Initializing	View alarms +	ap-south-1b
☐		i-0cd774afcfd666872	Running	t2.micro	2/2 checks passed	View alarms +	ap-south-1b
☐	MyNewInstance	i-047b683cf433b381f	Running	t2.micro	2/2 checks passed	View alarms +	ap-south-1a
☐		i-00760703ad97051fb	Running	t2.micro	2/2 checks passed	View alarms +	ap-south-1a

Select an instance

CloudShell

Feedback

© 2024, Amazon Web Services, Inc. or its affiliates.

Privacy

Terms

Cookie preferences